



GE HEALTHCARE

Technical Publications

GE Part Number 5343114
Revision A1

TEAL PDU Module in XGD2 Cabinet

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Operating Documentation

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IMPORTANT SAFETY INSTRUCTIONS

SAVE THESE INSTRUCTIONS – This manual contains important instruction for the XGD2 Power Distribution Unit (PDU) that must be followed during installation, operation, and maintenance.



⚠ DANGER



Electrical shock hazard. Opening enclosure exposes hazardous voltages. Refer service to qualified personnel.

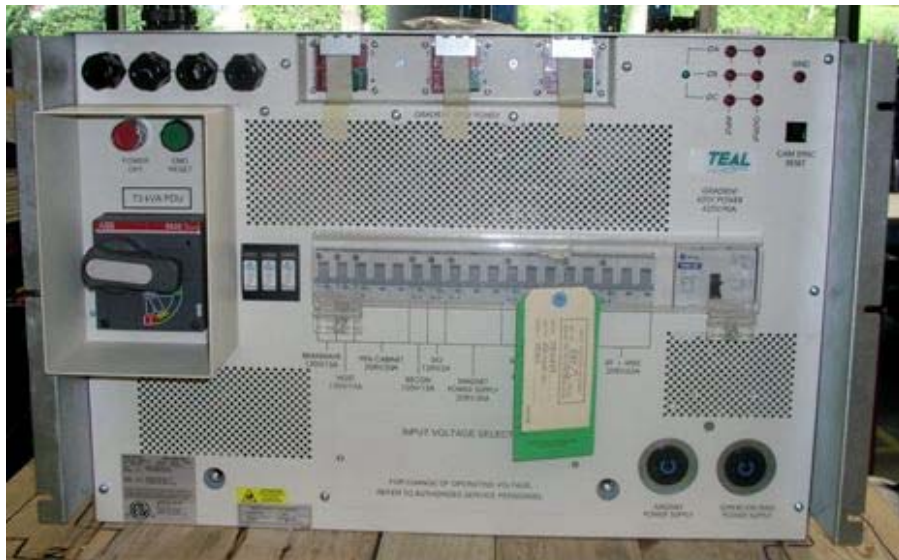
⚠ IMPORTANT

As standards, specifications, and designs change from time to time, please ask for confirmation of the information given in this publication.

1- INTRODUCTION

1.1 General Description

The XGD2 Power Distribution Unit (PDU) distributes power to various components of a General Electric Medical Systems MR system while providing surge suppression and electrical isolation for the power line. Three phase input voltage is selectable among 380/400/415/480 VAC at a rated frequency of 50/60 Hz. The power rating for this PDU is 73 kVA continuous and 97 kVA instantaneous. The PDU mounts into the GE 5173027 cabinet.



**XGD2 Power Distribution Unit
Illustration 1-1**

1-2 Scope

This manual describes procedures for installation and basic operation of the XGD2 Power Distribution Unit as well as procedures for failure diagnosis/isolation and installation of field replaceable components of the PDU.

2- SPECIFICATIONS

GE Specifications: 5343114PSP

GE Part Number
5343114

2.1 Electrical Specifications

Power Rating

73 kVA Continuous; 97 kVA Instantaneous

Input Voltage

380/400/415/480 VAC Delta.

Output Voltage

208/120 VAC WYE and 242/420 VAC Wye

Output Circuit Breaker and Connection Requirements

Breaker Size	Function	Ref. Desig	Sec Voltage	Wire to Phase	E-Stop	Customer Access	Connection Type
90A 3 Pole	Gradient 420V Power	CB2	420	ABC	Yes	Front	APP#1321G4, G5(3 Connectors)
63A 3 Pole	RF + MNS	CB3	208	ABC	Yes	Front	Breaker
20A 3 Pole	Pen Cabinet	CB4	208	ABC	Yes	Top Left Side	Breaker
15A 1 Pole	Brainwave	CB5	120	A	No	Top Left Side	Breaker
15A 1 Pole	Host	CB6	120	A	No	Top Left Side	Breaker
32A 3 Pole	Supercon Shim PS	CB7	208	ABC	No	Front	5 wire twist lock (Nema style L21-30R Hubbell) (receptacle front of cab)
32A 3 Pole	Magnet Power Supply	CB8	208	ABC	No	Front	5 wire twist lock (Nema style L21-30R Hubbell) (receptacle front of cab)
10A 2 Pole	Gradient 48V Control	CB9	208	BC	No	Top	7W2 D-Sub connectors
20A 1 Pole	Recon	CB10	120	B	No	Top Rear	NEMA 5-20R receptacle
2A 1 pole	SIO	CB11	120	B	Yes	Top Rear	NEMA 5-20R receptacle
1A 1 pole	CAM Sync Reset	CB12	6.7VAC	N/A	No	Front	N/A

Overload Protection

Input main circuit breaker (CB1). Output distribution circuit breakers.

Load Current

67 Amps total maximum continuous (208V)

68 Amps total maximum continuous (420V), 134A Peak 5 sec Max.

Transformer Impedance

Less than 2%

Efficiency

98% Typical

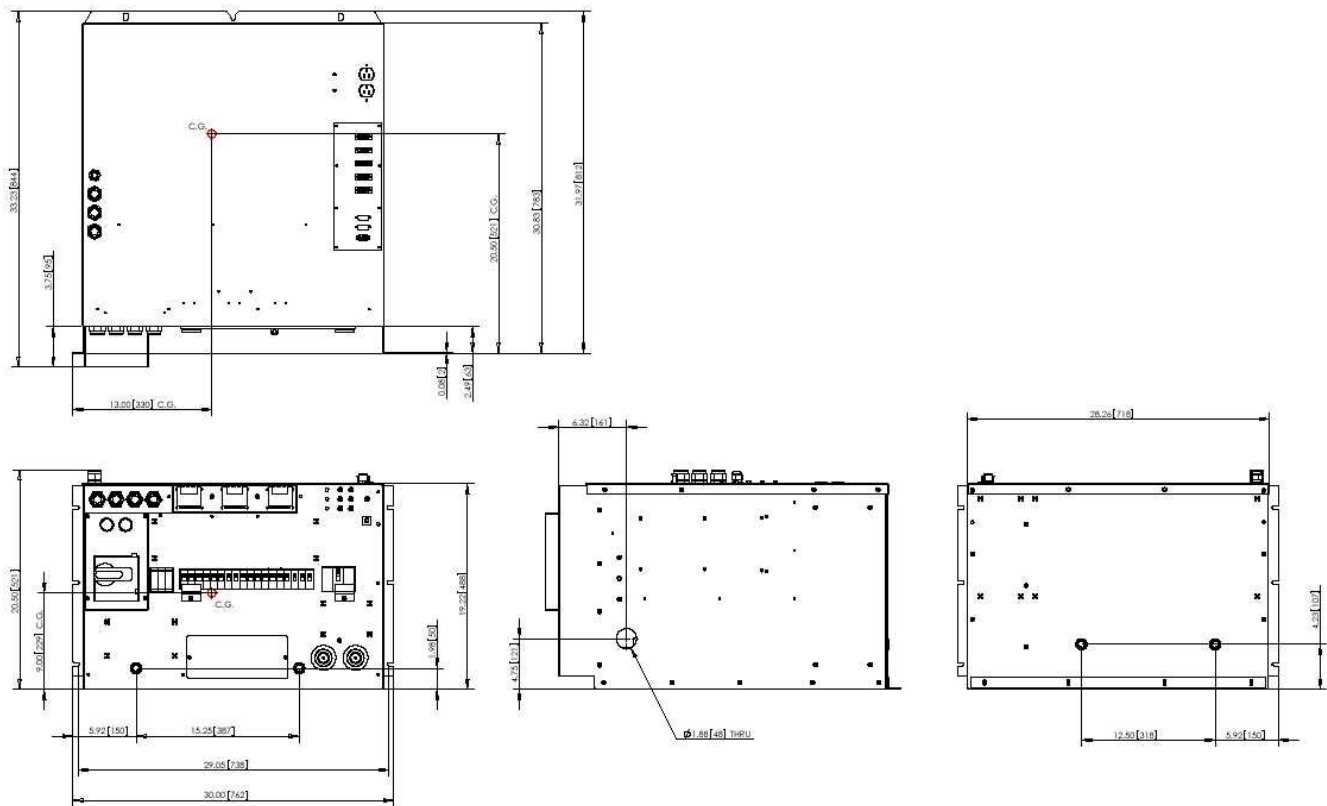
2.2 Environmental Specifications**Heat Dissipation**

4,076 BTU/Hr

2.3 Mechanical Specifications**Weight**

879 lbs. / 398 kg.

Dimensions



XGD2 Overall Dimensions
Illustration 2-1

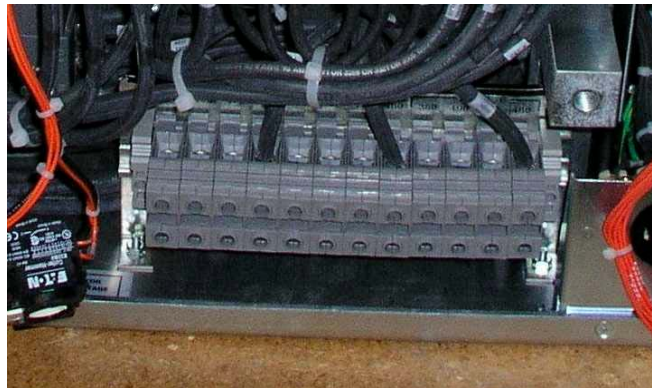
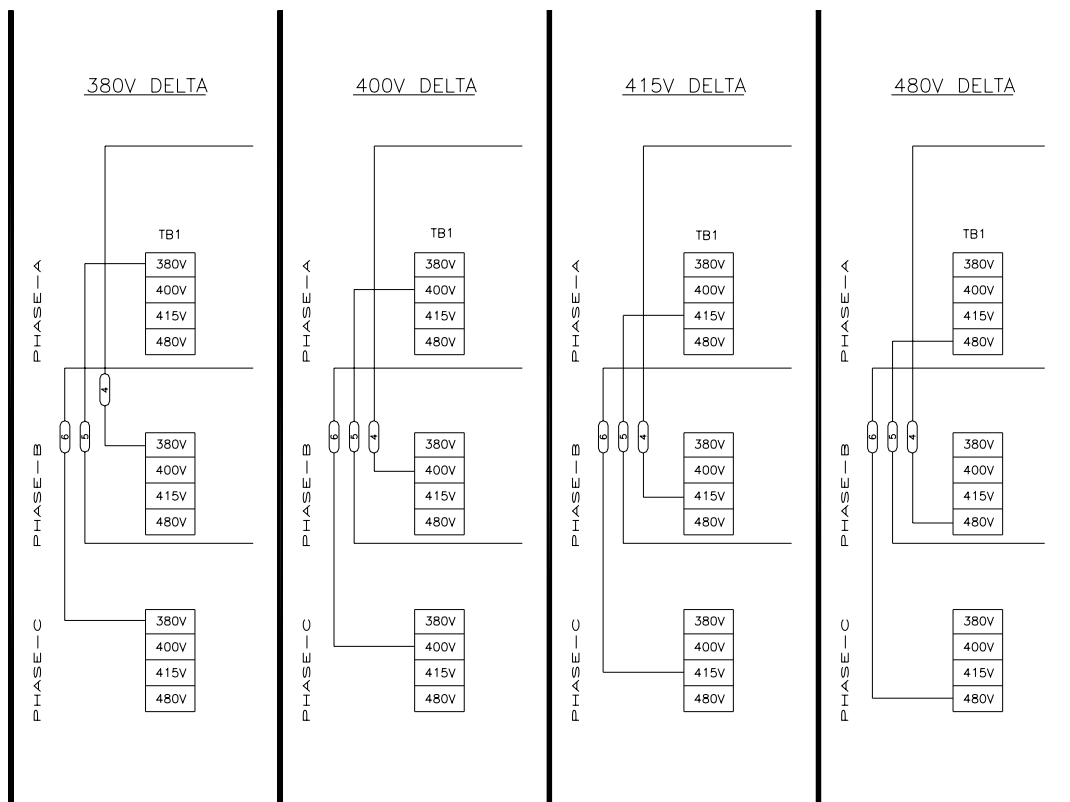
3- INSTALLATION AND OPERATION

3.1 Installation

3.1.1 Input Voltage Selection

Before connecting the input power cables, determine the nominal value of the input voltage. The PDU allows for nominal input voltages of 380, 400, 415, or 480 VAC (three phase). Input voltage selection is accomplished as follows:

1. On the Main Disconnect Panel (MDP), turn off the PDU circuit breaker, lock and tag appropriately.
2. Using a voltmeter or other voltage indicating device at the Main Input Terminal Board, check to be sure that no voltage is being applied to the input of the PDU.
3. Remove the small cover panels from the front panel of the PDU marked "INPUT VOLTAGE SELECT".
4. Illustration 3-1 shows the Input Voltage Selection terminal block (TB1). The illustration shows TB1 configured for 380V Delta. For each of the three phases, be sure the wire is connected to the proper terminal for the actual input voltage at the site and that the terminal screw is tight to make a good connection.

**INPUT VOLTAGE SELECT TERMINAL BLOCK (TB1)****Illustration 3-1****INPUT VOLTAGE SELECTION****Illustration 3-2**

3.1.2 Connection

Note the Cable Access Panel shown in Illustration 3-5 above the main circuit breaker. This panel provides strain relief cable clamps for the input and the various output cables. This panel may be removed from the PDU to provide easier access to the terminal blocks.

1. Route the cables through their respective cable clamps and connect cables to the appropriate terminals as shown.
2. Replace the Cable Access Panel and tighten the cable clamps.



For input cables, use wire sizes in accordance with local regulatory requirements for a 150A load

3.2 Operation

3.2.1 Main Input Circuit Breaker

The Main Input Circuit Breaker operating handle has three positions:

- OFF (as shown in Illustration 3-4),
- ON (vertical), and
- TRIPPED (slightly to the right of vertical). If the Main Input Circuit Breaker has been tripped, it is necessary to move the handle to the OFF position to reset it before turning it ON.



MAIN INPUT CIRCUIT BREAKER OPERATING HANDLE
Illustration 3-4

The Main Input Circuit Breaker may be locked in the OFF position. By pressing on the small triangle on the operating handle, the lockout tab is made accessible to a padlock.

3.2.2 Power Off

This pushbutton at the top of the front panel, when pressed, will immediately trip the Main Input Circuit Breaker, interrupting all power to the load.

3.2.3 EMO Reset

This pushbutton immediately to the right of the Power Off pushbutton will enable the Emergency Stop circuit.

3.2.4 TVSS Indicators

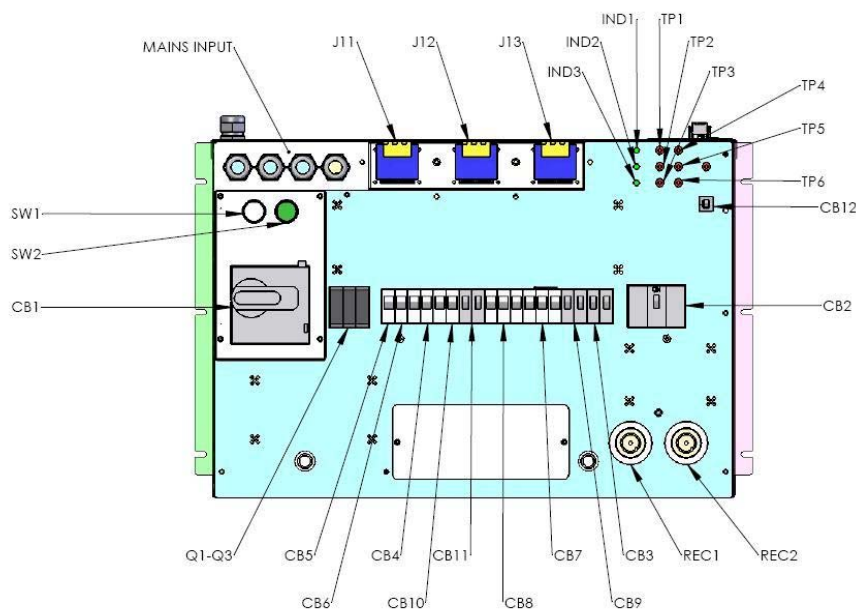
To the right of the Main Input Circuit Breaker are three TVSS (Transient Voltage Spike Suppressor). Any module that fails must be replaced to restore the PDU to operation with surge protection functioning. See Section 4-1.1, Replacement of a TVSS Module, for instructions for replacing a TVSS Module.

3.2.5 Phase Indicators

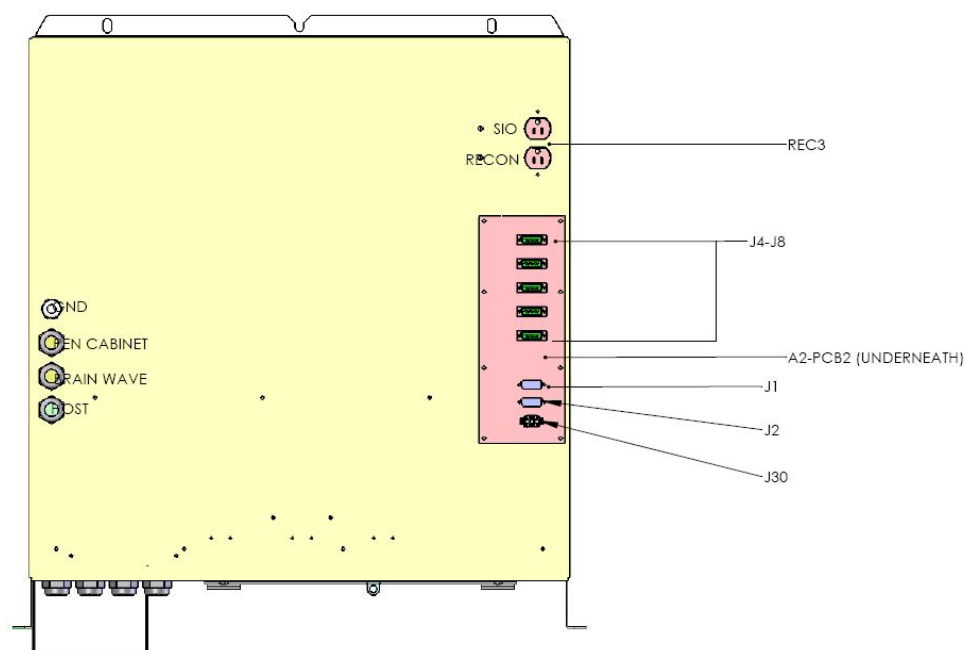
There are three Phase Indicator lights. These indicate that the transformer is energized and ready to deliver power to the load.

3.2.6 Input / Output Voltage Test Points

Near the Phase Indicators on the front panel are jacks into which can be plugged connectors for testing for input and output voltage. See Illustration 3-5. The voltage at these points are reduced by a factor of 100:1 from the actual voltage at the terminal blocks.



FRONT PANEL COMPONENT LOCATIONS
Illustration 3-5



TOP PANEL COMPONENT LOCATIONS
Illustration 3-6

4- FIELD REPLACEABLE UNITS (FRU'S)

LISTING OF FRU's
4-1

Item	Ref Designator	Description	GE P/N	TEAL P/N
1	A1	Control PC Board AUX Config	5343114-2	4990954
2	A1	Control PC Board PEM Config	5343114-3	4990955
3	Q1-Q3	TVSS Module	2301200-3	4990387
4	CON1	420V E-Stop Contactor	5343114-4	4990950
5	CON2	208V Distribution Contactor	5140621-3	4990781
6	PS1-PS2	48V Power Supplies	5140621-4	4990782
7	FAN1-FAN2	48V Fan with Tach	6140621-5	4990783

4.1 FRU Installation

4.1.1 Replacement of the TVSS Module

The operation of the TVSS Indicators is explained in Section 3-2.4, TVSS Indicators. If one (or more) of the indicators shows that a TVSS Module has failed, follow the steps below to replace the failed module and restore TVSS protection:

1. Turn off the PDU circuit breaker, lock and tag.
2. Using a voltmeter or other voltage indicating device, check the voltage at the Main Input Terminal Board to be sure that no voltage is being applied to the input of the PDU.
3. Pull out the failed TVSS Module(s) and plug in the replacement(s) as shown in Illustration 4-1.



TVSS MODULE REPLACEMENT
Illustration 4-1

4. Re-energize the PDU.
5. Verify that all TVSS Modules are good (no red flags displayed).

4.1.2 Removal of Front Panel to Access Additional FRUs (time: approx. 15 min)

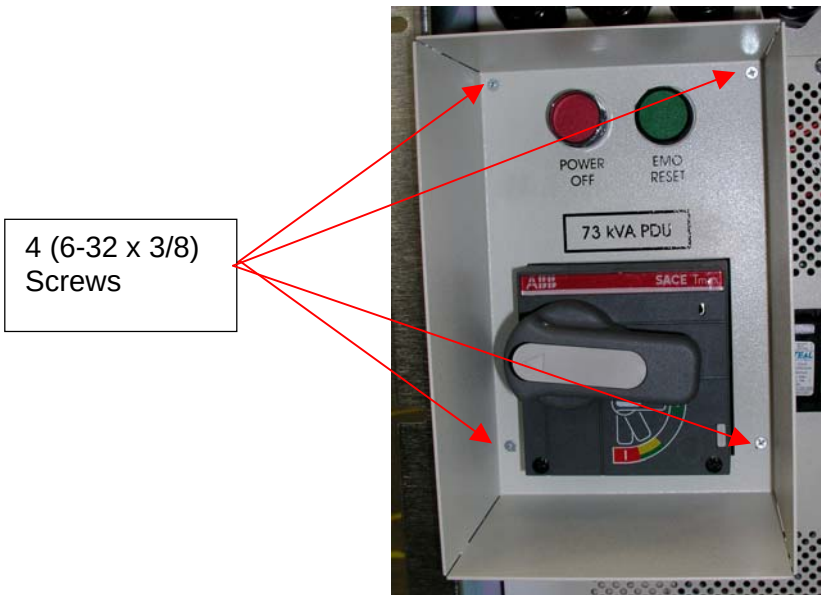
Tools needed: Phillips screwdriver or power screwdriver set, 11/32 open wrench.

1. Turn off the main input circuit breaker, lock and tag.

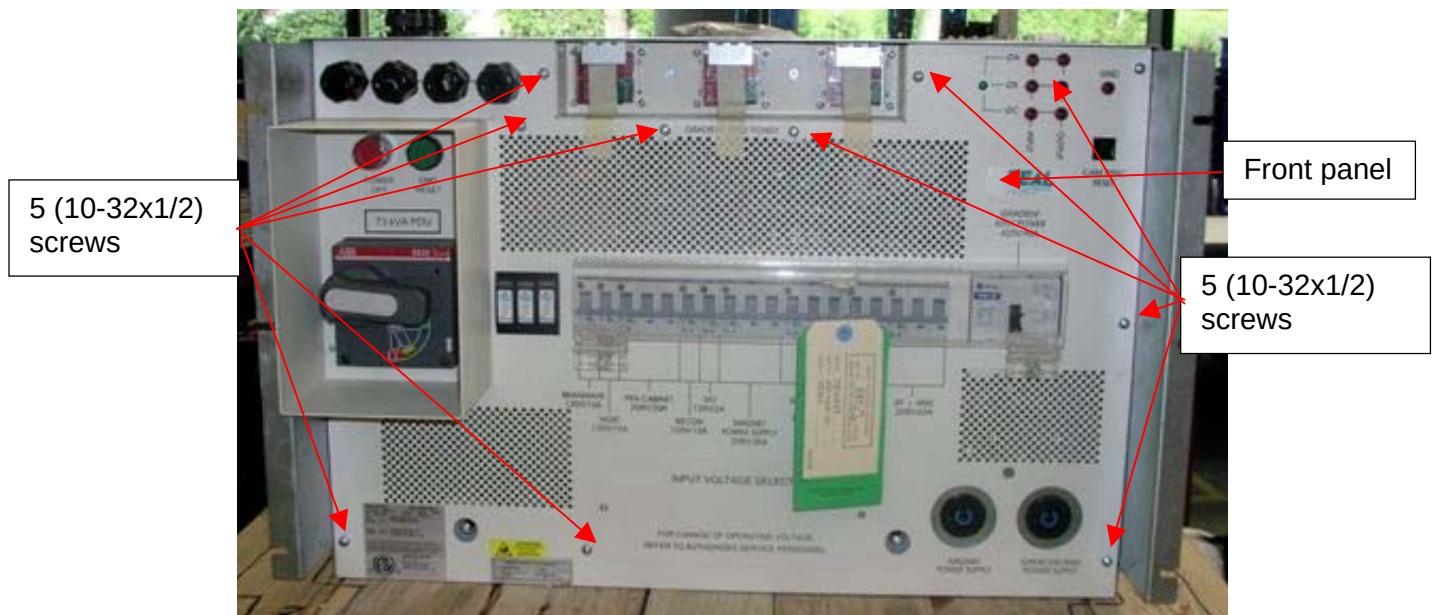
Input CB



2. Remove 4 (6-32 x 3/8) screws on the circuit breaker guard. Note: Do not discard screws, as they will be used to put back guard.



3. Remove 10 (10-32x1/2) screws on the front panel using Phillips screwdriver. Note: Do not discard screws, as they will be used to put back front plate.



4. Slightly pull the front panel and lean it toward the front.

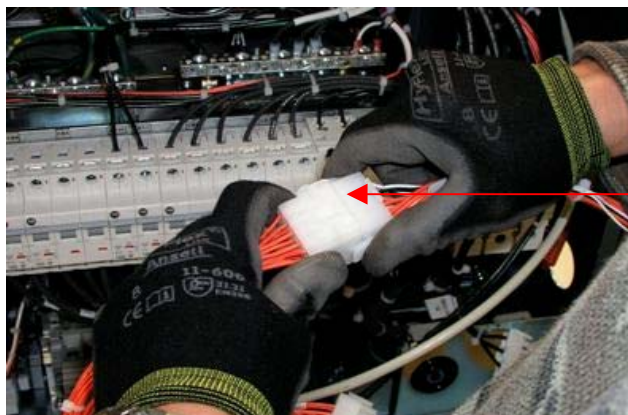


5. Disconnect SW1 and SW2 from the back of the front panel. Note: No tools will be required, turn tab counterclockwise and gently pull assembly apart.

SW1 and SW2



6. Disconnect P8 and J8 15Pin housing connector. Note: No tools required, press locking tab and pull connectors apart.



15Pin housing
connector (P8/J8)

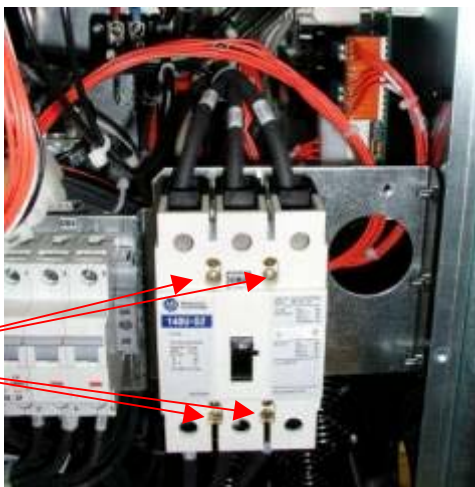
4.1.3 Replacement of the Control PC Board (time: approx. 35 min)

If the control board is deemed defective as per the Troubleshooting Guide in 4-1.8 follow the steps below to replace the control board. The control board is mounted on the inside of the side panel.

Circuit boards or assemblies with static sensitive devices will be packaged in accordance to prevent damage due to ESD (Electrostatic Discharge). Procedures to prevent ESD should be followed when handling static sensitive devices

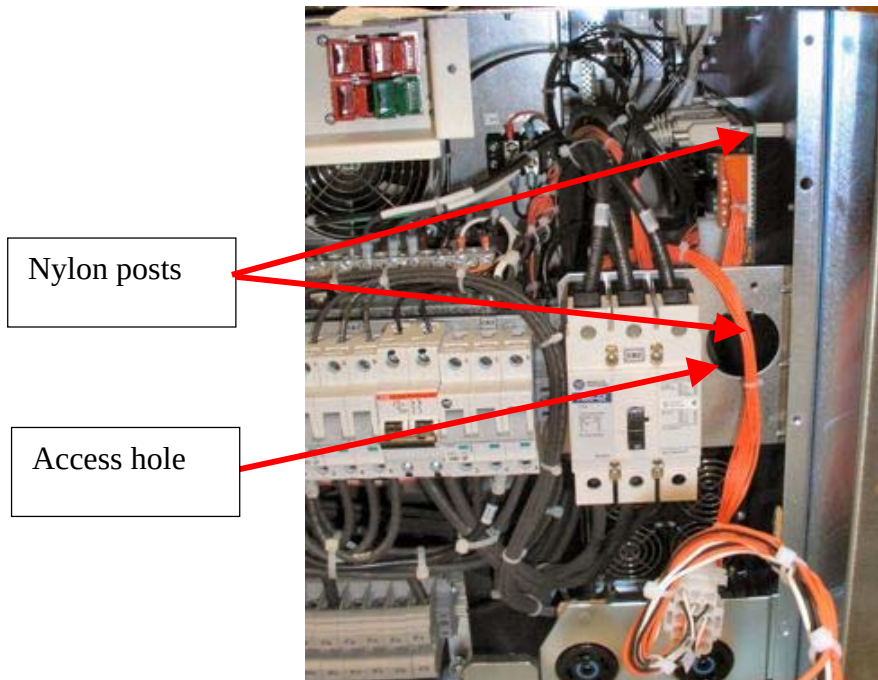
1. Remove the front panel as per 4.1.2
2. Remove the four mounting screws from the circuit breaker (CB2) and move the CB to the side.

Note: Do not discard screws, as they will be used to replace CB.

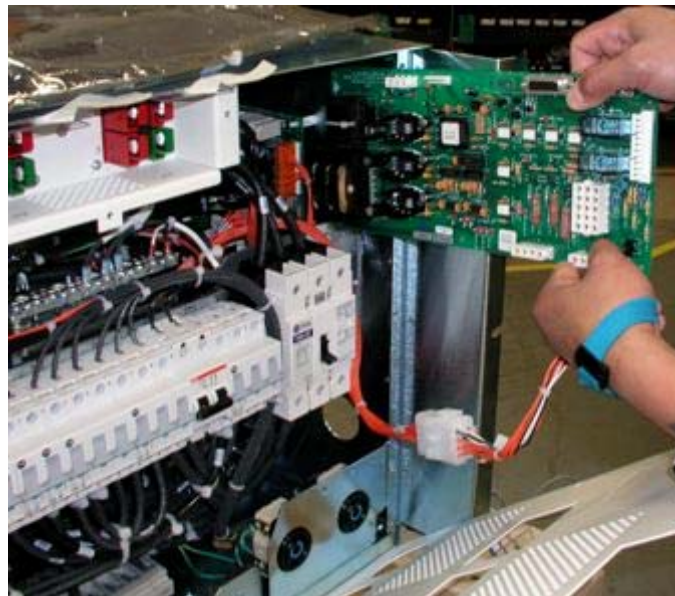


Remove 4
(8-32 x 3")
screws

3. Using the tool supplied with the PCB, pop the PCBA off the front two nylon posts. the lower post can be accessed through the port hole



4. With an ESD strap on, gently slide the PCBA out of the rear holders



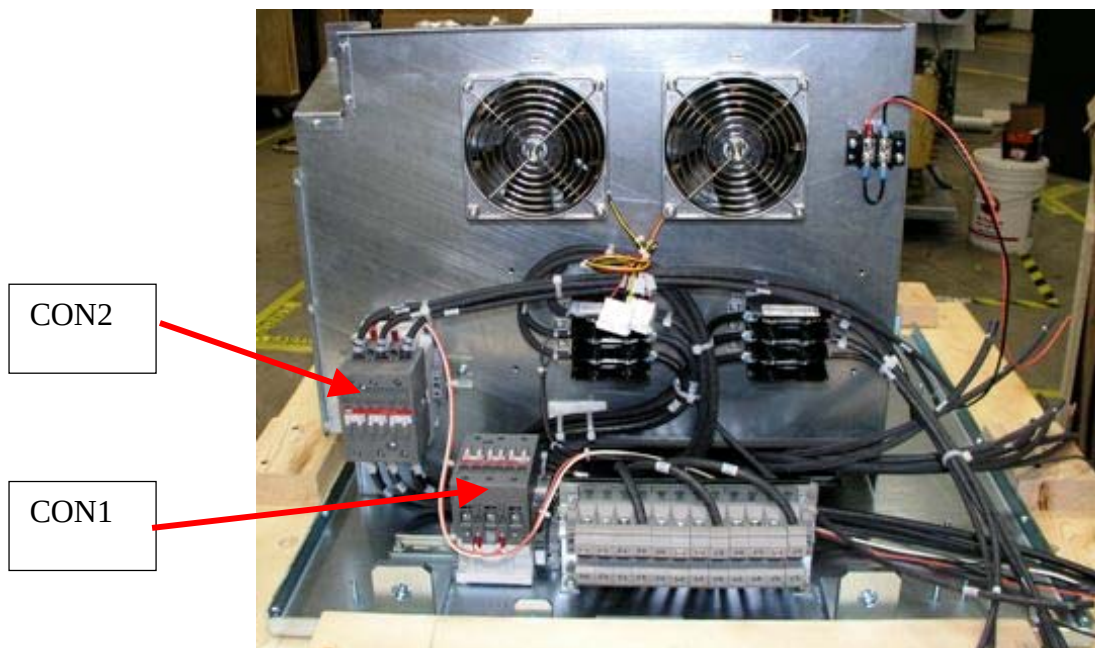
4. Disconnect cables P1, P2, P3, A1-P5, A1-P6, A1-P7, A1-P8. Mark the cables with a masking tape and marker to ensure correct position when re-connecting the cables.

4.1.4 Replacement of Contactors (CON1 or CON2) (time: approx. 30 min each after front panel removed)

If one or more of the contactors are deemed defective as per the Troubleshooting Guide in 4-1.8 follow the steps below to replace the contactors. The contactors are mounted behind the circuit breaker panel of the PDU located on the front panel.

Contactors or assemblies with static sensitive devices will be packaged in accordance to prevent damage due to ESD (Electrostatic Discharge). Procedures to prevent ESD should be followed when handling static sensitive devices.

Tools needed: Phillips screwdriver or power screwdriver set, slotted screwdriver, 11/32 open/closed or socket wrench.



1. Remove the front panel as per 4.1.2
2. Disconnect wires 97, 98 at the AUX of CON1 and wires 200 – 202 at T1-T3 of CON1. Make sure you mark the wires with a masking tape and marker.
3. Snap out CON1 from the mounting bracket and disconnect wire 148 – 150 at L1 – L3.
4. Disconnect wires 69/180 at A1 of CON1 and wire 147/181 at A2 of CON1. Make sure you mark the wires with a masking tape and marker.
5. Remove CON1 from unit and proceed to removing CON2.
6. Disconnect wire 151 at L1, 152 at L2 and 153 at L3 of CON2.
7. Snap out CON2 and disconnect wires 128/63 at T1, 65/129 at T2 and 130/66 at T3 of CON2.
8. Disconnect wire 180 at A1 and 181 at A2 of CON2.
9. Remove CON2 from unit and prepare to install the replacement parts. Replace old contactor with a new contactor. Re-connect the wires on CON1 and or CON2 based on schematic diagram. Tighten all connections. Torque specification for T1-T3, L1-L3 is 35 in. lbs. And torque specification for AUX and A1-A2 is 9 in. lbs.

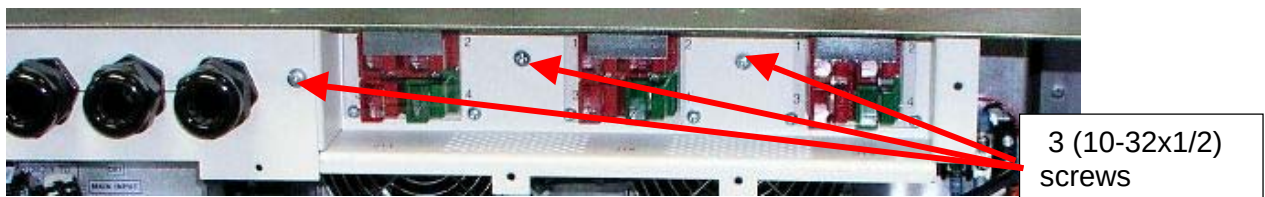
4.1.5 Replacement of Fans (FAN1 or FAN2)) (time: approx. 30 min after front panel removed)

If the fans are deemed defective as per the Troubleshooting Guide in 4-1.8 follow the steps below to replace the fans. The fans are mounted behind the circuit breaker panel of the PDU located on top of the unit.

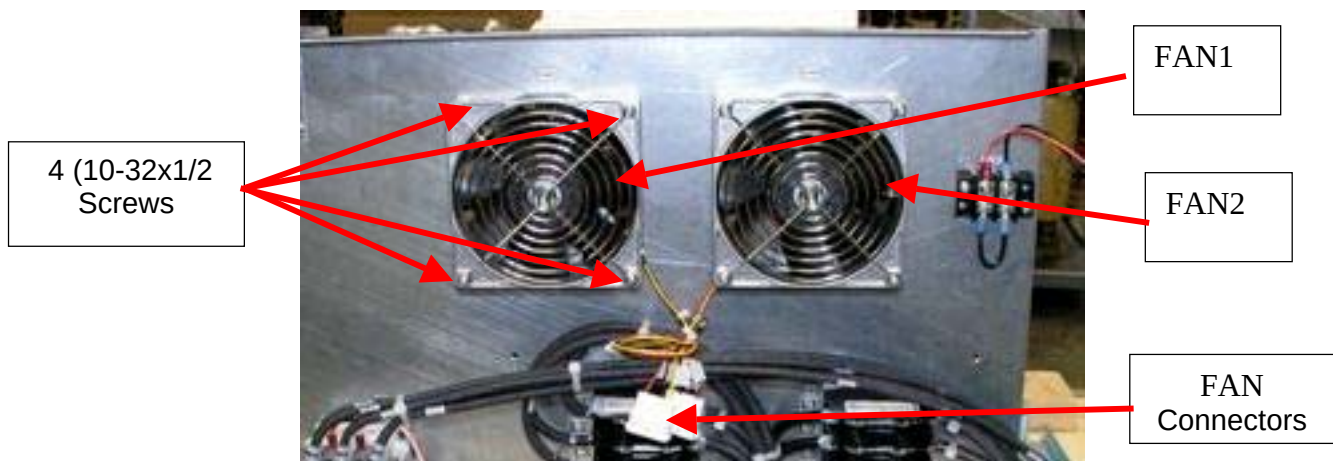
Fans or assemblies with static sensitive devices will be packaged in accordance to prevent damage due to ESD (Electrostatic Discharge). Procedures to prevent ESD should be followed when handling static sensitive devices.

Tools needed: Phillips screwdriver or power screwdriver set.

1. Remove the front panel as per 4.1.2
2. Remove 3 (10-32x1/2) screws at the Anderson assembly connector. Note: Do not discard screws, they will be used to replace assembly.



3. Remove fans (FAN1 and FAN2) by unscrewing 4 (10-32x1/2) screws using Phillips screwdriver. Note: Do not discard screws, they will be used to replace fan assembly.



4. Unplug fan connectors and remove fan assembly from unit.
5. Replace old fan assembly with FRU using the screws previously removed. Re-connect housing connectors P14 with J17 and P19 with J19.

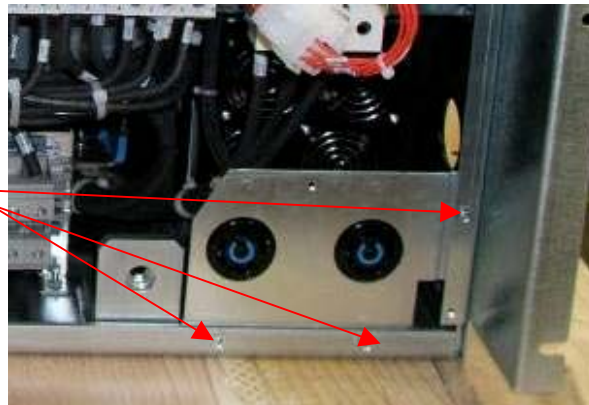
4.1.6 Replacement of Power Supplies (PS1 & PS2)) (time: approx. 30 min after front panel removed)

If the power supplies are deemed defective as per the Troubleshooting Guide in 4-1.8 follow the steps below to replace the power supplies. The power supplies are mounted behind the receptacle assembly of the PDU located on bottom of the unit.

Tools needed: Phillips screwdriver or power screwdriver set.

1. Remove the 3 (10-32x1/2) screws at the receptacle assembly. And move assembly out of the way.

3 (10-32x1/2)
screws

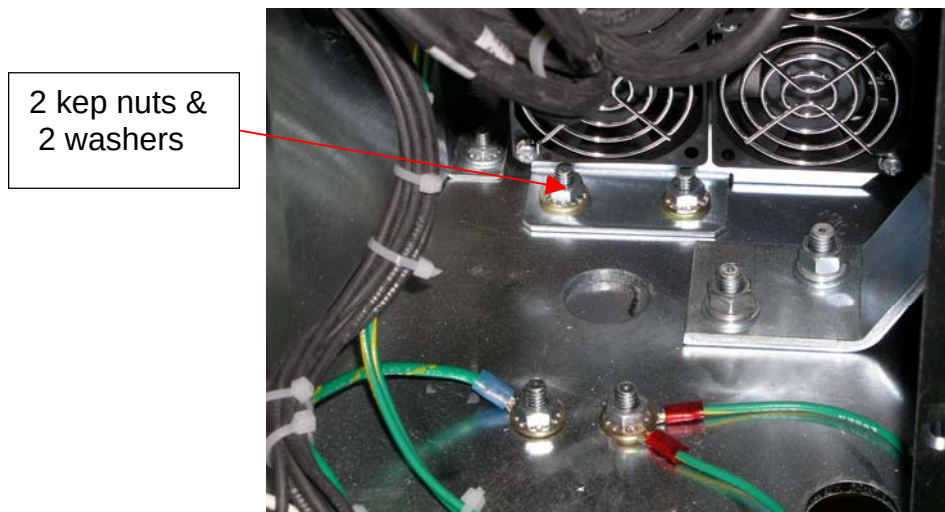


2. Disconnect wires 200, 201 & 202 from the bottom of CB2

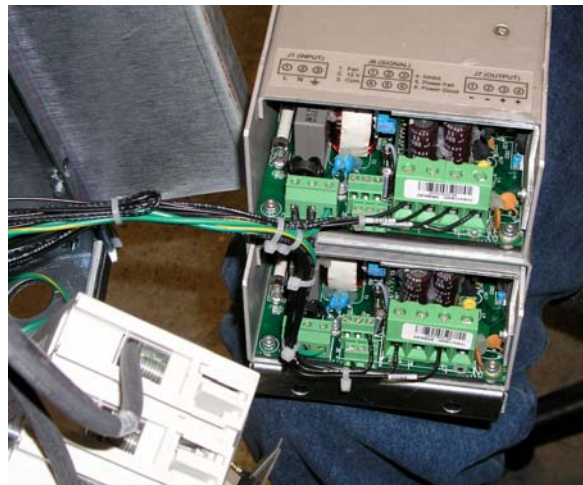
Wire 200, 201
& 202



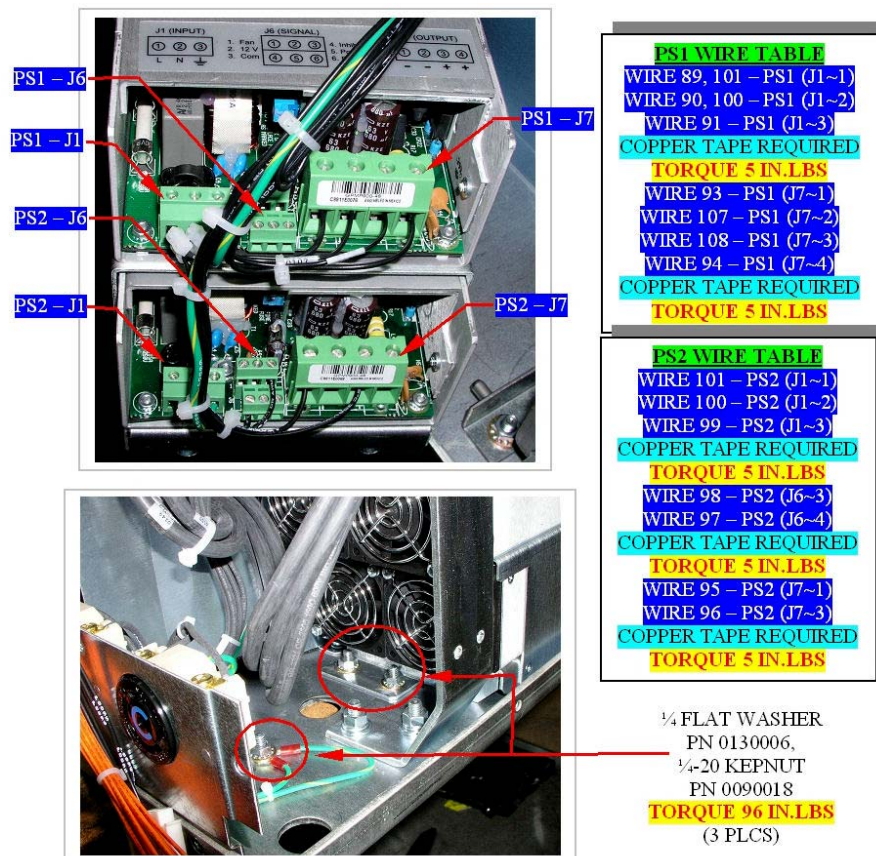
3. Remove 2 KEP nuts and 2 washers at the power supply assembly.



4. Carefully pull out the power supply assembly from the unit.
5. Disconnect ALL wires from the power supply assembly. Make sure you mark the wires with a masking tape and marker



6. Reconnect wires to the new power supply assembly per the instructions in step 8. Replace by power supply by performing these instructions in the reverse order.
7. Following are connection directions and torque specification requirements for replacing the power supply



Note: If a torque screwdriver is not available, when the torque value is listed as 5 in-lbs., tighten the screw until it is snug. When the torque value is listed as 96 in-lbs., tighten the screw/nut until it is firm.

4.1.7 Placement of New FRUs

1. To replace the FRUs in Sections 4.1.3 through 4.1.6 following the instructions in reverse order.
2. Use the schematic as needed to ensure placement of wires and/or cables.
3. To close the unit back up following the directions of Section 4.1.2 in reverse order.
4. Once unit has been closed up turn on Main Input circuit breaker.

4.1.8 Troubleshooting Guide

The following table lists fault conditions that may be encountered along with possible causes and solutions.

Troubleshoot Guide

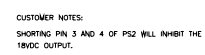
Table 4-2

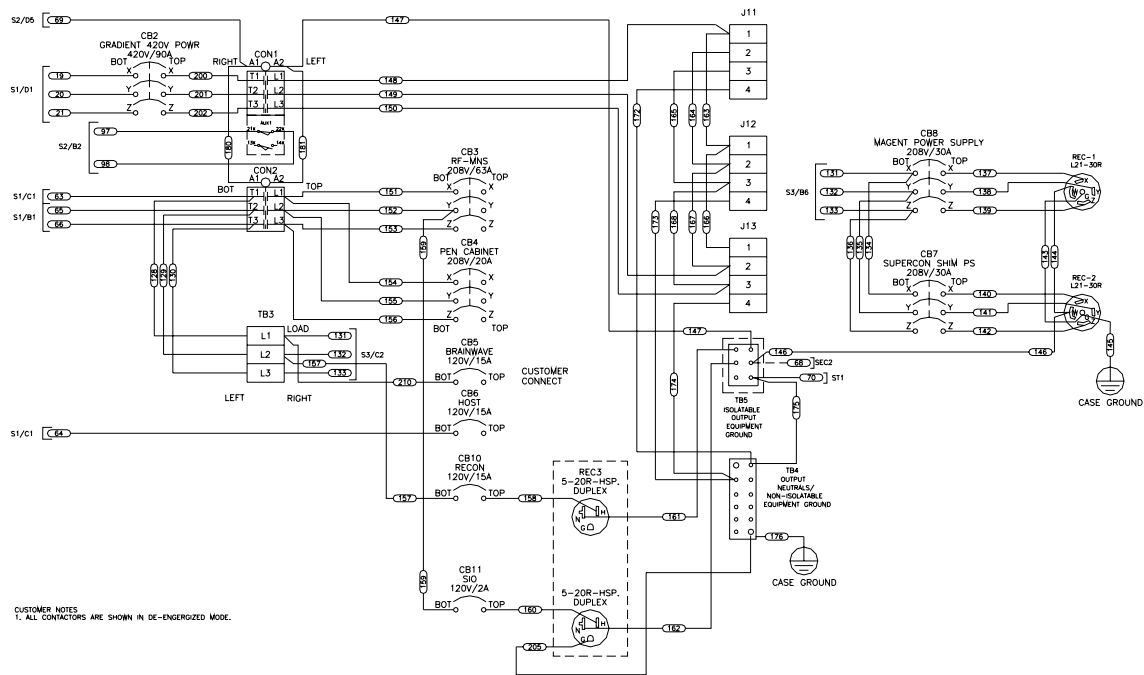
MODULE	SYMPTOM	CAUSE	SOLUTION
PDU	Main Input CB Trips	Over-temperature	1. Investigate cause of overtemperature 2. Allow unit to cool down. 3. Reset the circuit breaker
		Shorted Power Off Switch	1. Turn power off 2. Reset Main Circuit Breaker 3. Turn power on 4. If CB trips, replace PDU
	Distribution Breakers Tripping	Defective Circuit Breaker	1. Turn power off 2. Check if circuit breaker still trips (won't latch up)
		Overload	1. Check load by disconnecting cabinet from PDU whose circuit breaker is tripping, or use an Amp Clamp / DVM to check current draw. 2. Correct load problem.
	CON1/CON2 Contactor not operating	Defective Contactor	1. Ensure that the contactor CON1 is receiving 120V to the coils. If 120V is present, replace the contactor. 3. If 120V is not present, verify that CON2 is receiving 120V to the coils 4. If 120V is present, replace the contactor. 5. If 120V is not present, replace the PDU.
		Defective Control Board A1	1. Connect test plug and verify PDU operation. 2. If PDU operates with Test Plug, then replace the cable that connects to J2. 3. If PDU does not operate with Test Plug, then replace Control Board A1
	Indicators not lighting when power is applied	Input Circuit Breaker off	1. Turn Input Circuit Breaker, CB1, on
		Input Phase Missing	1. Check that other phase LED's are on 2. Measure at TP
		Light Burned Out	1. Check that other phase LED's are on. 2. Replace PDU
		Not Getting Proper Output	1. Measure input and output, at front panel test points, of each phase, from phase to phase (208 VAC: 208 VAC) and phase to ground (120 VAC: 120 VAC) 2. If difference is not a 100:1 ratio, replace PDU

Troubleshoot Guide
Table 4-2 (cont.)

MODULE	SYMPTOM	CAUSE	SOLUTION
PDU	Unit does not exit ESTOP mode (cannot turn on output contactor)	Control Board not receiving reset signal	1. Try resetting the unit by pressing SW1 on the front panel
		Control Board not receiving power	1. Confirm connection on the board 2. Confirm board is receiving 120V by measuring at P6-11(hot) and P6-9 (neutral)
	Fans not running	CB9 is OFF	1. Ensure CB9 is in the ON position
		PS1 Power supply not receiving power	1. Ensure CB9 is in the ON position 2. With CB9 on, measure for 120VAC at PS1 J1-1 and PS1 J1-2
		Defective power supply	1. With CB9 on and 120V present at PS1 J1, look for the green LED to be lit on the front of the power supply 2. If the green LED is not lit, then the power supply is likely defective, replace the power supply 3. If the green LED is lit, then measure for 48VDC at the power supply output PS1 J7 4. If 48V is not present, the likely the power supply is defective, replace the power supply
		Defective Fan	1. Measure for 48VDC at the fan connectors P4-1 & P4-3 and P5-1 & P5-3 2. If 48VDC is present, then replace the fan
	Voltage at one or more output missing	Output breaker in the OFF position.	1. Turn ON all output breaker
		Unit is still in ESTOP mode	1. Apply ESTOP reset signal
	Internal system AC line clock signal missing	CB12 open	1. Reset CB12 "CAM Sync Reset" from PDU front panel

SCHEMATICS





REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
A0			New Release
A1	5/28/10	R. Gregg	1. Changed "5-20" to "NEMA 5-20R" in section 2.1 2. Added TEAL FRU P/N's to table 4-1 3. Added sentence "The illustration shows TB1 configured for 380V Delta" in section 3.1.1 4. Corrected "PS3" to be "PS2" in section 4.1.6 title 5. Change "board" to "Control Board" and "Power Supply" to "PS1 Power Supply" in table 4-2 6. Removed the cause "Open from J3-2 to J3-3" and it's respective solutions from table 4-2, troubleshooting guide 7. Removed solutions 1-3 from cause "Defective Control Board A1" in table 4-2 troubleshooting guide 8. Rewrote the solution section of cause "PS1 Power Supply not receiving power" in table 4-2 troubleshooting guide 9. Added Symptom of "Internal system AC line clock signal missing." Add Cause of "CB12 open.". Add Solution of "Reset CB12 "CAM Sync Reset" from PDU front panel." to table 4-2 troubleshooting guide 10. Replaced component location illustrations, 3-5 & 3-6 with correct version drawings 11. Replaced "XGD" with "XGD2" in sections 1.1, 1.2, 2.3 and various headers 12. Added reference to CB12 in table in section 2.1 13. Corrected 4th instruction in section 4.1.3 to be numbered "4." in stead of "1."
A2	6/28/10	GEHC	Alternate torque values added by GEHC to section 4.1.6